INTERIM FEDERAL SPECIFICATION

HOT LINE STICKS

This Interim Federal Specification was developed by Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for hot line sticks with attachments for use in repair and maintenance of live, high voltage, aerial lines.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types. Hot line sticks shall be of the following types as specified (see 6.1):

Type I - Lift pole, conventional jaw, with swivel ring end butt.
Type II - Fuse puller and disconnect end.
Type III - Clampstick.
Type IV - Tie stick and fit-on head.
Type V - Switch hook stick with disconnect head.
Type VI - Universal pole with fit-on head.
Type VII - Tool, jumper holding.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issues in effect on date of invitation for bids, or request for proposals, form a part of this specification:

Federal Specifications:
QQ-S-781 - Steel Strapping, Flat.
PPP-B-636 - Box, Fiberboard.
PPP-S-760 - Strapping, Nonmetallic, (And Connectors).

Federal Standard:
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D-149-64 - Standards Methods of Tests for Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Uniform Freight Classification Rules:

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Ill. 60606.)

National Motor Freight Classification Rules:

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N. W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of sticks otherwise in accordance with this specification.

3.2 Material and design. Hot line sticks shall be of rigid fiberglass of high-impact resistance, and impervious to moisture. They shall be easy to clean and withstand the dielectric test specified in 4.4.1 without arc-over or visible damage to the tool. Each stick shall have passed a proof test by the manufacturer of 100 KV each ft. 5 minutes.

3.2.1 Tolerances. A tolerance of plus or minus 8 inches will be permitted on the overall length of the poles, exclusive of fittings.

3.2.2 Color. Hot line sticks shall be yellow, orange, red, tan or a combination of orange and black.

3.3 Type I, lift pole, conventional jaw, with swivel ring end butt. Type I lift pole shall be equipped with a head having jaws that can be tightened around a conductor. Head shall be nonferrous or corrosion-protected ferrous metal. Clamping screw shall be plated steel. The opposite end of the stick shall be equipped with a free swiveling butt ring for attaching to lever support or for supplemental arm brace.

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3.3.1 Lift pole shall perform the same function as the tool shown in figure 1, and shall be of the dimensions specified in table I.

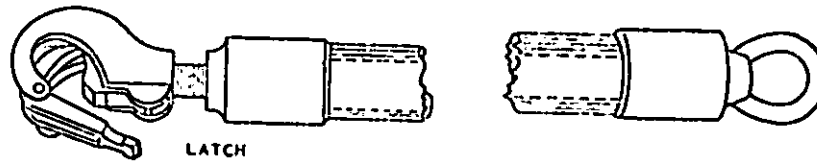


FIGURE 1. Type I, lift pole, conventional jaw, with swivel ring end butt.

TABLE I. Type I, lift pole, conventional jaw, with swivel ring end butt.

Pole diameter (min.)	Pole length (see 3.2.1)	Cable diameter accommodation size (max.)
Inches	Feet	Inches
2	10	1-1/8
2-1/2	16*	1-1/8

*16-foot poles may be supplied in two sections (see 3.3.2)

3.3.2 When 16-foot poles are supplied, they shall be furnished with splices or sleeves that make a connection as rigid and strong as the pole. Splices may be of steel or aluminum alloy.

3.4 Type II, fuse puller and disconnect end. The type II fuse puller shall consist of a hot line stick as specified in 3.2 and 3.2.1 equipped with an angle-type head or jaws. The jaws shall be designed to open or close when the operator turns the stick to the desired position. The head shall be of bronze or aluminum alloy. The jaws shall be padded or coated with plastic to provide a sure grip when puller is in operation. The head shall be designed to permit angle adjustment through a minimum of 90°. A button tip suitable for opening or closing disconnects or fuse clips shall be provided.

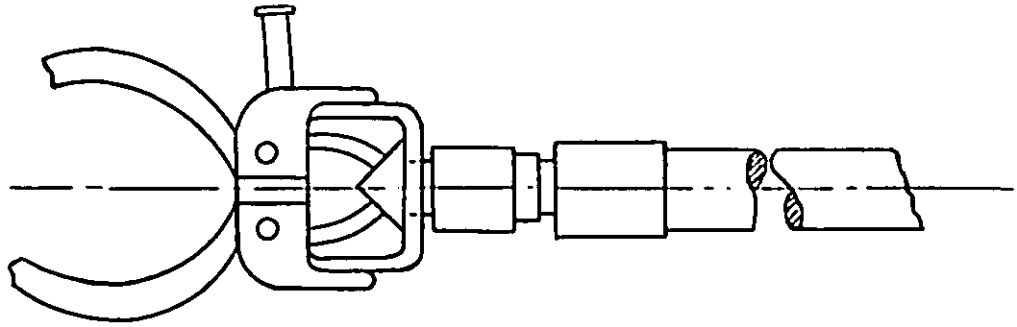


FIGURE 2. Type II, fuse puller and disconnect end.

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TABLE II. Type II, fuse puller and disconnect end.

Pole diameter (min.)	Pole length (see 3.2.1)	Fuse tube size	
		Min.	Max.
Inches	Feet	Inches	Inches
1-1/4	10	1/2	1-1/2
1-1/4	12	1-1/2	3
1-1/4	16	1-1/2	3

3.5 Type III, clampstick. Type III clampstick shall consist essentially of a fiberglass rod(s) equipped with an opening and closing hook, or clamp, on the outer end for applying tap and grounding clamps; opening fuse cutouts; and removing fuse tubes. Clamp, or hook, shall be heavy duty. An upper ferrule shall be fitted over the working end of the clampstick and enclose a rod and spring arrangement for opening and closing the hook or clamp. The outer end of the sleeve shall be slotted. The hook shall be designed to accept a 3/4-inch minimum outside diameter wire.

3.5.1 Safety latch. A latch(s) shall be located on the handle end of the clampstick to permit positive locking of the clamp in the shut position. Design may be the manufacturer's option, but it shall be of safety-type to prevent accidental release of the clamp jaws.

3.5.2 Pole hanger. Clampstick shall be provided with a pole hanger which may be adjusted to various locations along the clampstick.

3.5.3 The clampstick shall be similar to figure 3, and be of the dimensions specified in table III.

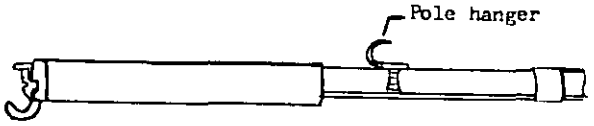


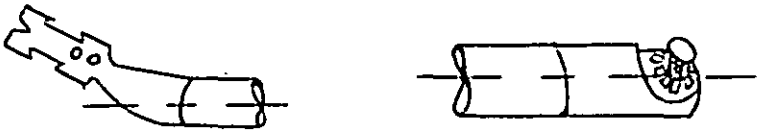
FIGURE 3. Type III, clampstick.

TABLE III. Type III, clampstick.

Pole diameter (min.)	Pole length (see 3.2.1)
Inches	Feet
1-1/4	6
1-1/4	8

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3.6 Type IV, tie stick. The type IV tie stick shall consist of a hot line stick (see 3.2 and 3.2.1), fitted with a fixed blade on one end as shown on figure 4, and a fit-on head on the opposite end (see figure 4). The blade shall be of bronze, steel, or heavy cast aluminum, and sharp-notched in a manner to permit working ties that do not have looped ends, and for making short bends in tie wire. The fit-on head on butt end of the stick shall be splined to interlock with splined fittings on universal tools. It shall be of brass, bronze, or aluminum alloy. A thumb, or slotted head, screw shall permit secure tightening of the connections. A tool hanger, or wire assistant, shall be fitted to the pole. A slip-on rubber hand guard shall be furnished with clamp for tightening on the stick. The tie stick shall be similar to figure 4. Pole shall be 1-1/4 inches minimum diameter, and 8 feet long (see 3.2.1).



Head

Butt end with fit-on head

FIGURE 4. Type IV, tie stick.

3.7 Type V, switch hook stick. The type V switch hook shall consist of a hot line stick as specified in 3.2 and 3.2.1, equipped with a disconnect head as shown on figure 5. Stick shall be 1-1/4 inches minimum diameter, and 6 feet long (see 3.2.1), with plain butt end as shown on figure 2. Head shall be of bronze or other suitable material.

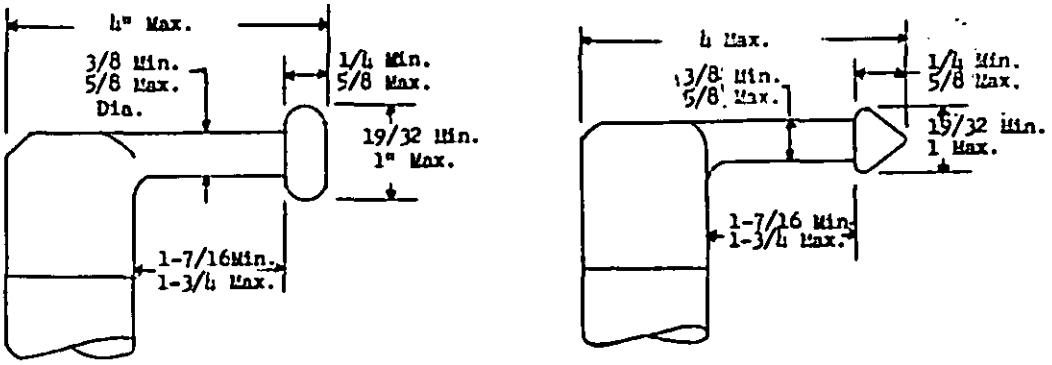


FIGURE 5. Type V, switch hook stick with disconnect head.
(Alternate style heads shown).

3.8 Type VI, universal pole with fit-on head. Type VI universal pole with fit-on head shall consist of a hot line stick as specified in 3.2 and 3.2.1, equipped with a fit-on head as specified in 3.6. Stick shall be equipped with two rubber skirts to protect the lineman's hands and reduce hazards of working in rain or snow. The skirts shall fit the stick snugly, and be shaped to deflect water streams that might run down the stick while in service. Stick shall be 1-1/4 inches minimum diameter and 8 feet long (see 3.2.1). Butt end shall be plain as shown on figure 2.

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3.9 Type VII, tool jumper holder. Type VII jumper holding tool shall consist of a hot line stick fitted with a clamp head. The upper part of the head shall have an extended beak for securely holding the conductor. It shall retain the conductor in the jaws. It shall be opened and closed by means of a separate control mounted on or in the handle so twisting on the main handle does not affect jaw closure. The jaws shall be shaped and formed to prevent the tool from walking off the conductor during tightening operation. The head shall be of copper, bronze, or aluminum alloy. The hot line stick shall be in accordance with 3.2 and 3.2.1, and equipped with a hanger as specified in 3.5.2.

3.9.1 The tool, jumper holder, shall be similar to figure 6, and of the dimensions and capacity specified in table V.

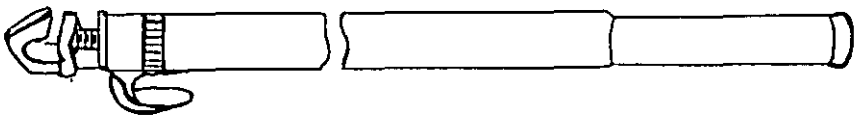


FIGURE 6. Type VII, tool, jumper holding.

TABLE V. Type VII, tool jumper holder.

Pole diameter (min.)	Pole length (see 3.2.1)	Conductor range
Inches	Feet	
1-1/4	4	6 Sol. to 336.4 MCM ACSR

3.10 Marking. Each hot line stick shall be plainly and permanently marked with the name of the manufacturer, or with a trademark of such known character that the source of supply may be readily determined. In addition, each stick shall be marked to indicate dielectric strength.

3.11 Workmanship. Workmanship shall be first class in every respect. The tool shall be free from cracks, burrs, or fins that might impair serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table VI.

TABLE VI. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 each	II	4.0	Defects per 100 units	4.3.1
Dimensional examination	1 each	S-4	2.5	Defects per 100 units	4.3.2
Testing	1 each	S-4	2.5	Defects per 100 units	4.4.1
Packaging and preservation	One container	S-2	4.0	Defects per 100 units	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, construction, workmanship, and marking.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Test.

4.4.1 Dielectric test. After immersion in water for 24 hours at a room temperature of 73° plus or minus 3° F., the surface of the stick shall be wiped dry and a one-foot section subjected to 100,000-volt a.c. test voltage for five minutes using test method ASTM D 149-64. The tool shall meet the requirements specified in 3.2.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table VII. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

TABLE VII. Classification of preparation for delivery defects

Examine	Defects
Materials	Any nonconforming component, component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as container flaps loose, container bulged or distorted.
Contents	Number of items per shipping container not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level B or C, as specified (see 6.1).

5.1.1 Level B. Each hot line stick shall be packaged in a snug-fitting fiberboard box conforming to PPP-B-636, class domestic, style optional. When hot line sticks are furnished in two sections, the two matching sections shall be fastened together in at least two places with a tape of the type normally used for the purpose.

5.1.2 Level C. Hot line sticks shall be packaged in accordance with the suppliers' normal practice in a manner which will protect the item from damage during shipment to the first receiving activity.

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5.2 Packing. Packing shall be level B or C, as specified (see 6.1).

5.2.1 Level B. Six packaged hot line sticks shall be strapped together with steel strapping conforming to QQ-S-781 or nonmetallic strapping conforming to PPP-S-760. Straps shall be placed approximately six inches from each end of the bundle and spaced between the end straps not to exceed 36 inches apart.

5.2.2 Level C. Hot line sticks, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the requirements of the Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable to the mode of transportation.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type and size required (see 1.2 and applicable tables).
(c) Selection of applicable level of preservation, and packing (see 5.1 and 5.2).
(d) Special markings required (see 5.3.1 and 5.3.2).

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

GGG-H-001347(GSA-FSS) - Hot Line Sticks

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

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